

Project Proposal

Sentiment Analysis and Rating Prediction on Hotel Reviews

6CS012

Artificial Intelligence and Machine Learning

Herald College Kathmandu

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1. Research Question

Can a Recurrent Neural Network (RNN) or a Transformer-based model accurately predict numerical hotel ratings (1–5) by analyzing the semantic content of customer reviews from TripAdvisor?

2. Title

A Performance Analysis of Deep Learning and Transformer Architectures for Multiclass Sentiment Classification of Hotel Reviews.

3. Dataset Description

Source

The dataset consists of 20,491 hotel reviews extracted from TripAdvisor, specifically curated for sentiment analysis and rating prediction tasks.

Dataset Composition

Number of Classes: 5 (Ratings 1, 2, 3, 4, and 5)

Total Number of Reviews: 20,491

Data Split: The data will be divided into Training, Validation, and Test sets (e.g., 70% Train, 15% Val, 15% Test)

Features:

- **Review:** Textual feedback provided by guests
- **Rating:** Numerical target variable (1–5)

Collection And Preparation Method

As the dataset is sourced from TripAdvisor, the following preparation steps will be applied:

Text Preprocessing: Conversion to lowercase, removal of punctuation, special characters, and stopwords to reduce noise.

Tokenization and Embedding: Transforming text into numerical sequences using techniques such as Word2Vec, or RNN model embedding layer.

Padding and Truncation: Ensuring all input sequences have a uniform length for model processing.

Data Splitting: Implementing a stratified split to ensure each rating class is represented proportionally in the training, validation, and test sets.

Challenges

Class Imbalance: Reviews often lean toward very positive (5) or very negative (1) ratings, which may bias the model against middle-range ratings (2–4).

Sarcasm and Context: Natural language often contains sarcasm or nuanced context that can be difficult for baseline models to interpret correctly.

Variable Review Length: Long, detailed reviews compared to very short ones require effective sequence handling (like attention mechanisms) to capture relevant sentiment.

4. Model Pipeline

We will be building three models to evaluate performance:

- **Baseline Model:** A simple Recurrent Neural Network (RNN) with a trainable Embedding layer.
- **Deeper Architecture with Regularization:** A Long-Short Term Memory (LSTM) with a trainable Embedding layer.
- **Fine-Tuning a Pre-Trained Model:** A Long-Short Term Memory (LSTM) with pre-trained Word2Vec embeddings.

5. Preliminary Data Analysis

Class distribution

Visualizing the frequency of each rating (1 through 5) to identify potential imbalances.

Rating Counts:

Rating

1 1421

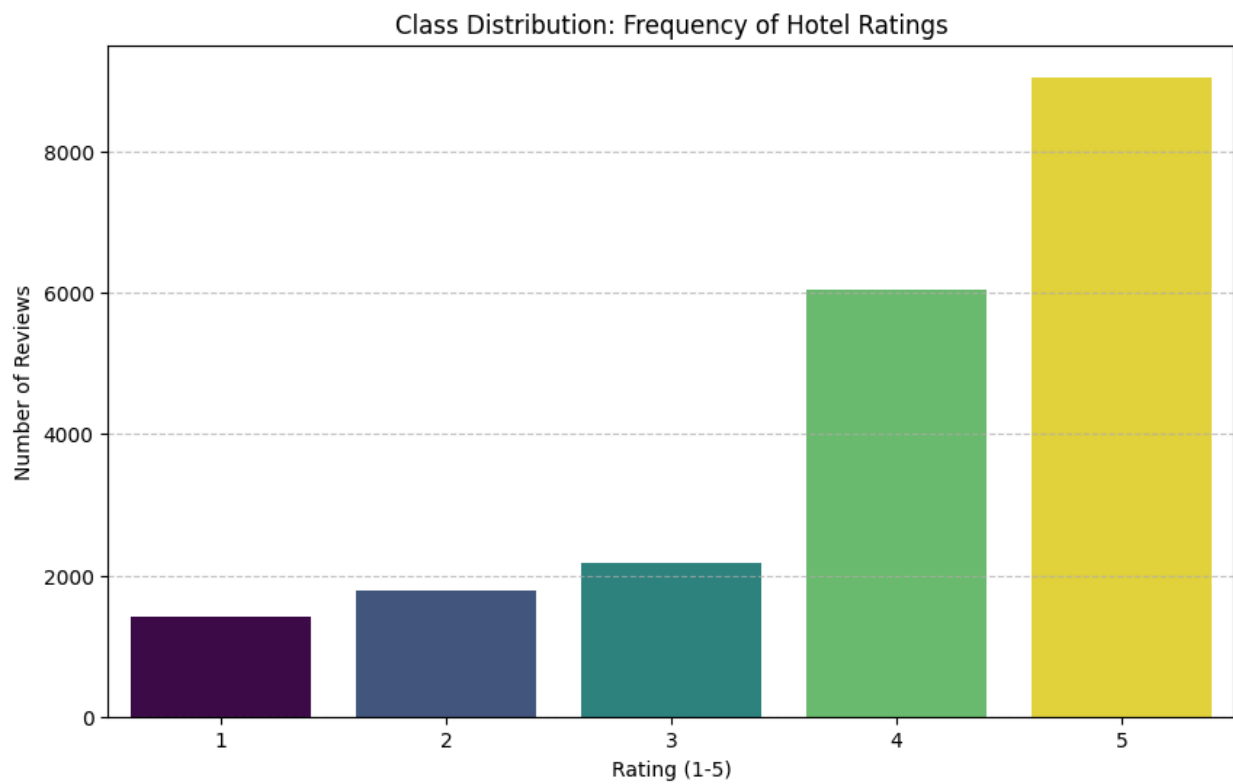
2 1793

3 2184

4 6039

5 9054

Name: count, dtype: int64



Sample Visualization

We will display sample reviews for each rating class to understand the linguistic differences between a "1-star" and a "5-star" review. For example:

```
-----  
SAMPLE REVIEWS PER RATING  
-----
```

```
--- Rating 1 ---
```

```
Sample: horrible customer service hotel stay february 3rd 4th 2007my friend picked hotel monaco appealing website onl
```

```
--- Rating 2 ---
```

```
Sample: ok nothing special charge diamond member hilton decided chain shot 20th anniversary seattle, start booked sui
```

```
--- Rating 3 ---
```

```
Sample: nice rooms not 4* experience hotel monaco seattle good hotel n't 4* level.positives large bathroom mediterrane
```

```
--- Rating 4 ---
```

```
Sample: nice hotel expensive parking got good deal stay hotel anniversary, arrived late evening took advice previous t
```

```
--- Rating 5 ---
```

```
Sample: unique, great stay, wonderful time hotel monaco, location excellent short stroll main downtown shopping area,
```